Cloud Architectures

Anandha Gopalan
(with thanks to Teo Yong Meng)

axgopala@nus.edu.sg

resource organization

Outline

Resource Organisation

Recall: Essential Characteristics NIST 2011

1. On-demand Self-service
2. Broad Network Access
3. **Resource Pooling**
4. **Rapid Elasticity**
5. Measured Service

NUS National University of Singapore

In Response to ...

- Fluctuation in cloud consumer's demand over time – **number of jobs**, and **resource demand within a job**
- Fluctuation in available cloud resources – cloud provider provides services to many cloud customers

Achieve elasticity or scaling

Achieve balanced utilisation of cloud resources

How is this done in practice?

Workload Distribution

Service Load Balancing

Resource Pooling

Dynamic Scalability

Elastic Resource Capacity

application level

resource level

Workload Distribution Architecture

Aim

Distribute consumer's workload over available cloud resources – use horizontal scaling

Key Component

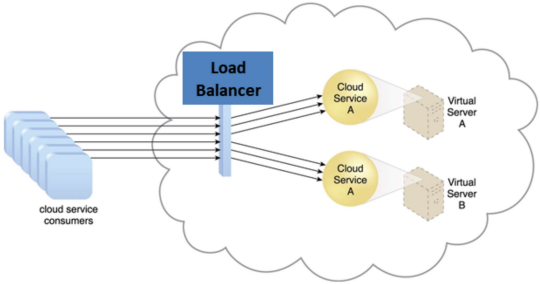
Load Balancer – distributes workload to reduce over-utilisation or under-utilisation of resources

horizontal scaling

NUS National University of Singapore

Workload Distribution Architecture

E.g., Amazon Elastic Load Balancing Service



Automatically distributes incoming application traffic

Achieve greater levels of fault tolerance for applications

Seamlessly provides the required amount of load balancing capacity needed

Load Balancer

AWS

Uses elastic load balancing ⇒ automatically distributes incoming traffic across multiple targets

Routes traffic only to healthy targets

Different Load Balancers

- **Application Load Balancer**
- Network Load Balancer
- Gateway Load Balancer
- Classic Load Balancer

network level - like roads and cars

gateway level - you want the right traffic on your roads

classic - for backward compatibility

Serves as a single point of contact for clients ⇔ contains one or more **listeners**

Listener

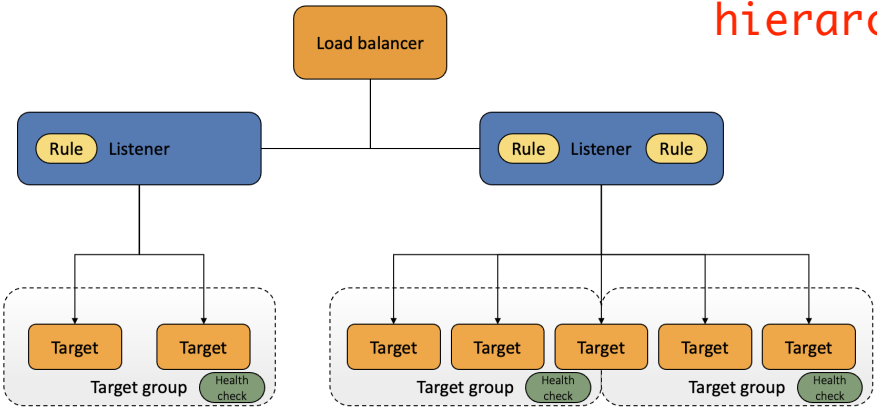
- Checks for requests from clients using the protocol and port that you configure
- User-defined rules routes requests to its registered targets (in target groups)
- Rules consist of **priority**, one or more **actions**, and one or more **conditions**
- A default rule is **necessary**

Steps

1. Load Balancer receives a request
2. Request is evaluated and appropriate rule is applied
3. Select a target from the target group for the rule action

protocol - the kind of request
port - where is the listener waiting

hierarchy



https://docs.aws.amazon.com/elasticloadbalancing/latest/application/introduction.html

Service Load Balancing Architecture

Specialised variation of the workload distribution architecture

Key Feature

Redundant cloud service deployments are created with a load balancer to dynamically distribute workloads

Resource pool $\Leftrightarrow$ **Duplicate** cloud service

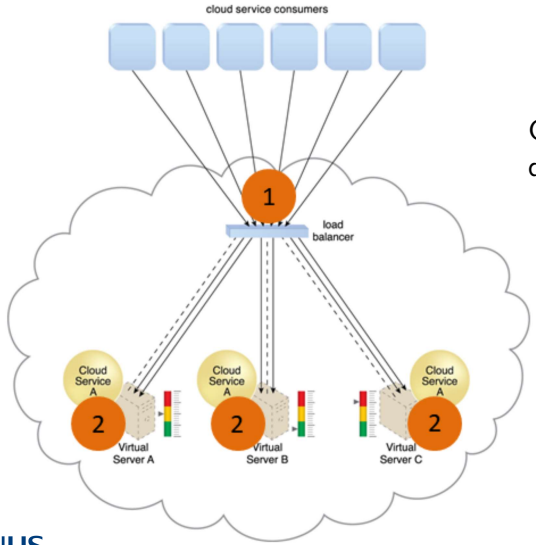
Load Balancers

Independent of cloud services and their host servers

Built-In to cloud service as part of application or server environment

NUS National University of Singapore

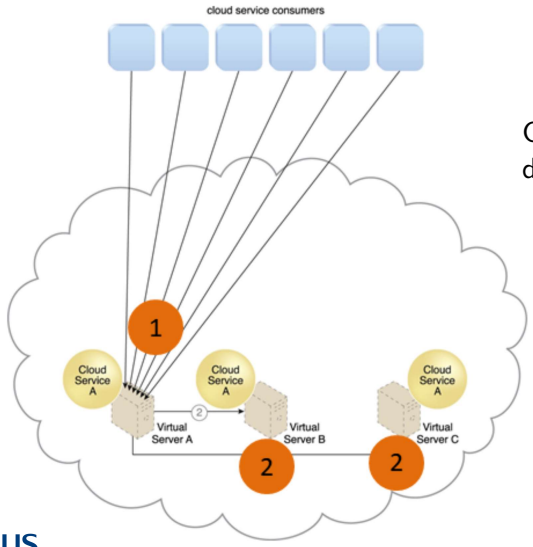
Independent Load Balancer



Cloud service A is **replicated** and runs on different virtual servers

1. Load Balancer intercepts requests from cloud consumers and forwards them to the virtual servers
2. Workload processing is horizontally scaled

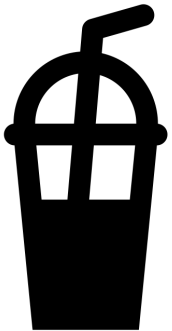
Built-In Load Balancer



Cloud service A is **replicated** and runs on different virtual servers

1. Cloud consumer's requests are sent to cloud service A on virtual server A
2. Built-in load balancing logic means that requests are distributed to virtual servers B and C

Take a Break







NUS National University of Singapore

Resource Pooling Architecture

Aim

Aggregate cloud resources of different types to serve diverse needs of cloud consumers

Key Feature

Automatically maintain identical IT resources to ensure they remain **synchronised**

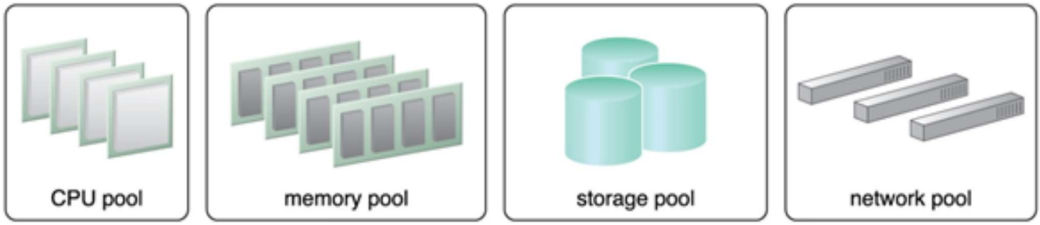
Common Resource Pools

- Server pool (Physical/Virtual)
- CPU pool
- Memory pool
- Storage pool
- Network pool

NUS National University of Singapore

Resource Pooling Architecture

Dedicated Pools



Example

Dedicated pool with 4 **sub-pools** with CPUs, memory, storage, and network devices

Issue!

Becomes complex $\Rightarrow$ multiple pools for specific consumer or application

Resource Pooling Architecture

Create hierarchical structure of parent, sibling, and, nested pools to meet diverse requirements

Sibling Resource Pool

Are drawn from physically grouped IT resources and not spread out across data centers

co-located resources

Isolated from one another so each cloud consumer is given access to its respective pool

Nested Resource Pool

Larger pools divided into smaller pools with same type of IT resources

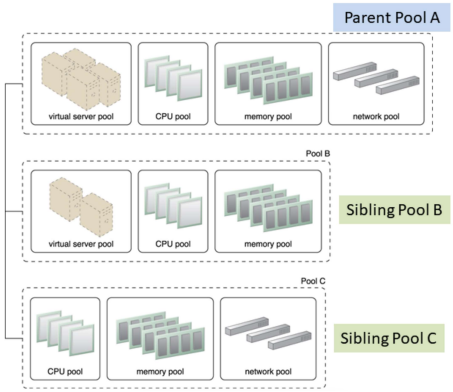
Used for assigning resource pools to different departments and groups within the same organisation

template

NUS National University of Singapore

Sibling Resource Pools

Example

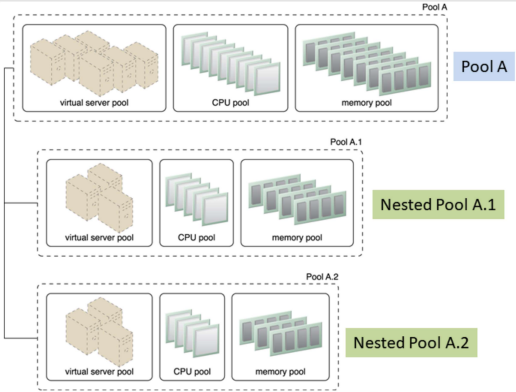


Pools B and C are sibling pools of Pool A

from same datacenter

Nested Resource Pools

Example



Pools A.1 and A.2 are nested pools of Pool A

same template, only different numbers

Dynamic Scalability Architecture I

Aim

Enable variable resource utilisation to meet **usage demand fluctuations**

Key Feature

Dynamic allocation of resources from resource pools based on predefined scaling conditions

NUS National University of Singapore

Dynamic Scalability Architecture II

Common Methods

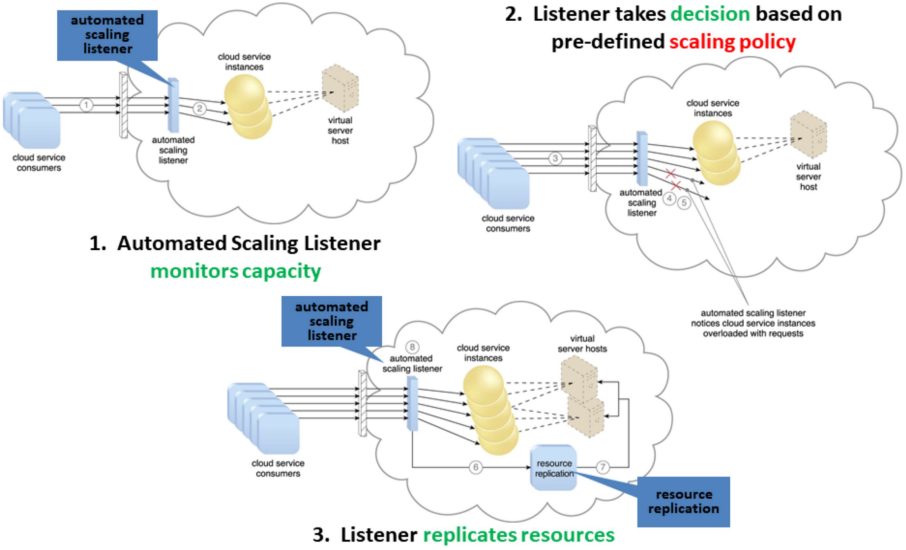
- Dynamic **Horizontal** scaling – **same** resource instance scaled dynamically with fluctuating workloads (e.g., replicating resources)
- Dynamic **Vertical** scaling – scale processing capacity of a **single IT resource** (e.g., increase memory dynamically when virtual server is loaded)
- Dynamic **Relocation** – resource relocated to host with larger capacity (e.g., faster storage device with better I/O capacity)

```
relocation -> move to entirely new machine
```

NUS National University of Singapore

Horizontal Dynamic Scalability

Example



Elastic Resource Capacity Architecture

Aim

Dynamic provisioning of virtual servers

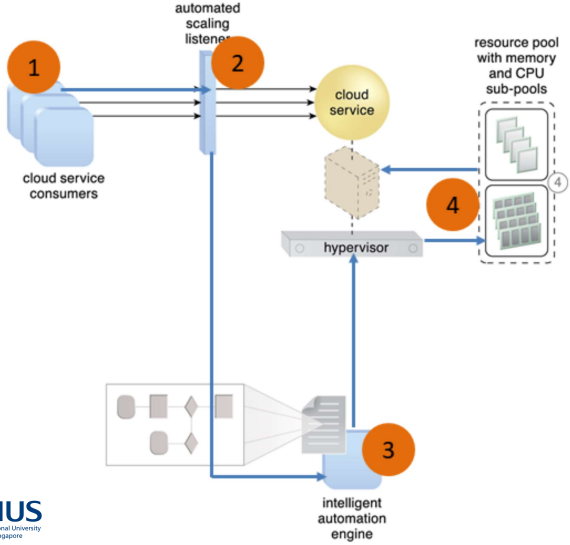
Key Features

- Fluctuating processing requirements
- Allocates and reclaims CPUs, RAMs, etc.
- Run-time virtual server is monitored
- Additional resources leveraged before capacity threshold reached
- Virtual server and IT resources are vertically scaled
- Uses Intelligence Automation Engine



Elastic Resource Capacity Architecture

Example

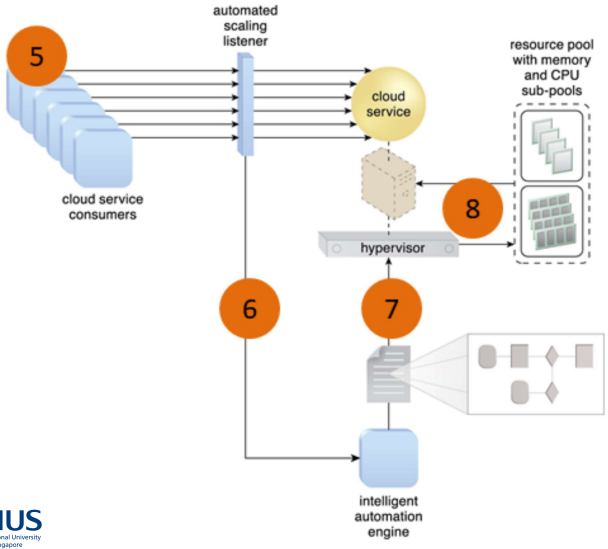


1. Cloud consumers send requests to a cloud service
2. Requests are monitored by an Automated Scaling Listener
3. Intelligent Automation Engine deployed with workflow logic
4. Workflow logic is capable of notifying the resource pool using allocation requests

NUS National University of Singapore

Elastic Resource Capacity Architecture

Example



5. Cloud consumer's requests increase dramatically
6. Automated scaling listener signals the Intelligent Automation Engine to execute the script
7. Script runs workflow logic that signals the hypervisor to allocate more resources from the resource pools
8. Hypervisor allocates additional resources to handle the increased workload

NUS National University of Singapore

Scaling On-Premise Provisioning

Aim

Scale on-premise IT resources into the cloud whenever capacity thresholds have been reached

Example

Cloud Bursting Architecture

Cloud Bursting Architecture

Definition

Burst Out – to meet higher usage demand, dynamic scaling of on-premise resources to cloud-based resources

Definition

Burst In – on lower demand, revert back to on-premise and release cloud resources

Cloud services are redundant, pre-deployed and inactive until cloud bursting

Features Required

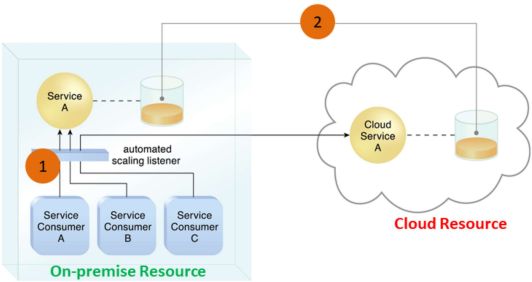
Automated Scaling Listener

Resource replication – resources on cloud and on-premise are **synchronised**

NUS National University of Singapore

Cloud Bursting Architecture

Example



Automated Scaling Listener monitors the usage of on-premise service A

1. When usage threshold is exceeded, Service Consumer C is redirected to Service A's redundant implementation on the cloud
2. Resource replication system ensures the state management databases are synchronised

Next Week

Resource Hosting and Datacenter

References

Chapter 11, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Continuous Feedback



https://forms.cloud.microsoft/r/ZJxTf2wFjr

NUS National University of Singapore